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NaYF₄:Er³⁺,Yb³⁺/SiO₂ Core/Shell Upconverting Nanocrystals for Luminescence Thermometry up to 900 K

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Abstract

The rapid development of nanomaterials with unique size-tunable properties forms the basis for a variety of new applications, including temperature sensing. Luminescent nanoparticles (NPs) have demonstrated potential as sensitive nanothermometers, especially in biological systems. Their small size offers the possibility of mapping temperature profiles with high spatial resolution. The temperature range is however limited which prevents the use in high temperature applications as *e.g.* nano-electronics, thermal barrier coatings and chemical reactors. In this work we extend the temperature range for nanothermometry beyond 900 K using silica coated NaYF₄ nanoparticles doped with the lanthanide ions Yb³⁺ and Er³⁺. Monodisperse *ca.* 20 nm NaYF₄:Yb,Er nanocrystals were coated with a *ca.* 10 nm silica shell. Upon excitation with infrared radiation bright green upconversion (UC) emission is observed. From the intensity ratio between ²H_{11/2} and ⁴S_{3/2} UC emission lines at 520 nm and 550 nm, respectively, the temperature can be determined up to at least 900 K with an accuracy of 1 to 5 K for

1 silica coated NPs. For bare NaYF₄:Yb,Er NPs the particles degrade above 600 K. Repeated thermal
2 cycling experiments demonstrate the high durability and reproducibility of the silica coated nanocrystals
3 as temperature probes without any loss of performance. The present results open avenues for the
4 development of a new class of highly stable nanoprobe by applying a silica coating around a wide
5 variety of lanthanide-doped NPs.
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13 The broad range of potential applications for nanomaterials continues to stimulate research in the
14 nanoregime where the size dependent physical and chemical properties are important in realizing
15 technological breakthroughs.¹⁻⁸ One research area involves the development of detection techniques
16 with sufficient spatial resolution enabled by the small size of nanomaterials. Temperature is an
17 important parameter, but is not easy to measure with sufficient resolution, *i.e.* at the (sub)micrometer
18 scale. In order to measure temperature accurately, several techniques have been developed. These
19 techniques can be divided into three different categories; electrical,⁹ mechanical¹⁰ and optical
20 techniques.^{11,12} Most electrical techniques combine the relation between temperature and resistance,
21 voltage, conductivity or electrical capacity with Atomic Force Microscopy (AFM). Mechanical
22 techniques revolve around an AFM tip composed of two metals and the temperature dependent contact
23 potential between these metals. Although a high resolution can be reached with these techniques, the
24 techniques are limited to surfaces. Furthermore, the techniques require contact with the sample and can
25 therefore initiate artificial heat fluxes.
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44 Optical techniques rely on changes in the absorption or emission spectrum induced by temperature
45 changes and do allow for remote temperature sensing. A promising method is luminescence
46 thermometry.^{13,14} Luminescence thermometry exploits the temperature-dependence of the spectral
47 position, bandwidth, intensity, polarization or lifetime of emission. However, most of these optical
48 properties can also depend on other parameters, such as concentration of luminescent centers, spectral or
49 intensity shifts induced by other factors besides temperature or fluctuations in the excitation power or
50 spectral alignment. Fluorescence intensity ratio thermometry (FIRT) is a ratiometric measuring
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technique and consequently does not depend on variations in sample concentration, alignment or excitation power. This makes FIRT the ideal technique to measure temperature, since it does not require an internal standard.

FIRT is an active and expanding field of research. Most of the research presently focusses on *in vivo* thermometry for bioimaging.^{15,16} Research has for example shown that FIRT can be applied for cancer diagnosis.¹⁷ In cancer, rapid mitosis leads to increased cell temperatures. These increased cell temperatures can be monitored at an earlier stage than with conventional techniques for cancer diagnosis. The temperature probes should have a high accuracy in the physiological range. The range and the accuracy of the temperature measurements cannot be optimized simultaneously. Consequently, the highly sensitive temperature probes used for cancer diagnosis have a very limited measuring range (below 350 K).

Many applications – *e.g.* temperature sensing in nano-electronics,^{18,19} nanophotonics,^{20,21} thermal barrier coatings^{22,23} and chemical reactors^{24,25} – require a much broader temperature range compared to bioimaging. FIRT can be a promising technique in a wider temperature range. However, this requires the temperature probes to be robust and photostable up to elevated temperatures. Ln^{3+} -doped materials are ideal candidates as temperature probes due to their sharp emission lines (easy detection against a broad background), high thermal quenching temperatures of their emission and high photostability. The Ln^{3+} -doped materials can be organic clusters, metal/organic frameworks, glasses or (nano)crystals.²⁶⁻²⁹ The use of lanthanides in inorganic hosts is favored over organometallic materials, mostly because of the higher (thermal and photo) stability of the inorganic host and the lower energies of vibrations (phonons) in close proximity to the Ln^{3+} ion. It is well known that the main mechanism for quenching of the 4f-4f emission of Ln^{3+} is multi-phonon relaxation, a process in which energy is transferred from a Ln^{3+} excited state to vibrational excited states (heat) of the surrounding matrix. The relaxation rate depends exponentially on the number of phonons that needs to be excited. As a result, multi-phonon relaxation has a much lower probability in inorganic hosts which have low energy phonons (*ca.* 600 cm^{-1} for fluorides and oxides) compared to organic materials where large energy gaps can be bridged by a

small number of high energy vibrations (*e.g.* 3000 cm⁻¹ and 3500 cm⁻¹ for C-H and O-H vibrations, respectively).³⁰

Based on these advantages, both doped glasses^{31,32} and doped crystalline materials³³⁻³⁵ have been investigated as functional materials for remote temperature sensing using FIRT. However, most of the works focus on bulk materials, with limited spatial resolution and no possibility to measure temperature in stable dispersions. Furthermore, research into both bulk and nanomaterials has led to temperature probes in a range up to 700 K.^{32,33,35} However, some applications require much higher temperatures, even up to 1500 K,³⁶ which cannot be measured with the currently available temperature probes. To measure temperature with sufficient spatial resolution and at elevated temperatures, new temperature (nano)probes need to be developed.

In this work, we report the application of NaY_{0.8}Yb_{0.18}Er_{0.02}F₄ nanocrystals (NCs) and NaY_{0.8}Yb_{0.18}Er_{0.02}F₄/SiO₂ core/shell (NaYF₄@SiO₂) nanocrystals as upconverting (UC) nanothermometers. The monodisperse core nanoparticles show temperature dependent UC luminescence up to 600 K. The luminescence drops at higher temperatures due to oxidation of the ligands and coalescence of NCs. To overcome this problem, a silica coating has been grown around the core NCs, resulting in core/shell NCs. These NaYF₄@SiO₂ NCs show stable temperature dependent UC luminescence up to 900 K. Repeated cycling between 300 K and 900 K shows that the durability of the NaYF₄@SiO₂ NCs is excellent. The accuracy of the measurements is also high, with standard deviations of 1K and 5K below and above 750 K, respectively. The present results show that NaYF₄ doped with Yb³⁺ and Er³⁺ can be used as a temperature probe up to 900 K after encapsulation in a protective silica shell. The broader temperature range, durability of the temperature probes and the non-invasiveness of FIRT makes this system ideal for measuring temperatures in a wide variety of systems and remote temperature sensing applications.

Structural Characterization. To determine the size, shape and crystallinity of both the NaYF₄ NCs and NaYF₄@SiO₂ NCs, we used different characterization methods such as transmission electron microscopy (TEM), energy dispersive x-ray spectroscopy (EDX) and powder x-ray diffraction (XRD)

measurements. Figure 1 shows TEM images of the prepared NCs and Figure S1 (Supporting Information (SI)) shows the corresponding EDX spectra and XRD diffractograms.

Figure 1a shows that the synthesis yields monodisperse NCs of 21.7 ± 1.1 nm. The individual NCs are well separated due to the steric effects of the ligand layer, which suggests that the NCs can form a colloidal stable dispersion. These monodisperse NCs form hexagonally ordered domains, which is illustrated with the Fourier Transform in the inset. The overgrowth with silica yields monodisperse $\text{NaYF}_4@\text{SiO}_2$ NCs with an unaltered core and a uniform shell of 10.0 ± 0.7 nm, which are charge stabilized by surface silanol groups.

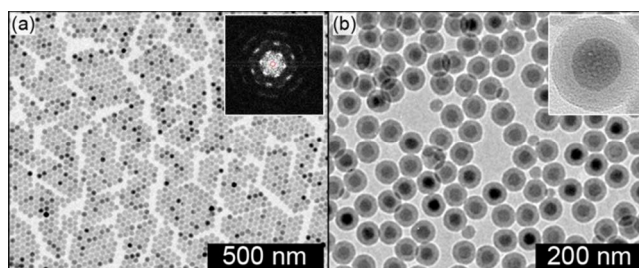


Figure 1. TEM micrographs of the prepared NaYF_4 NCs (a, including inset with a Fourier Transform) and $\text{NaYF}_4@\text{SiO}_2$ NCs (b). The average diameter of the core NCs is 21.7 ± 1.1 nm and the silica shell is 10.0 ± 0.7 nm.

The elemental composition of the NCs was determined by EDX (Figure S1a and b, SI). From the observed peaks, it can be concluded that Na, Y, F, Er and Yb are all present in the NaYF_4 NCS and $\text{NaYF}_4@\text{SiO}_2$ NCs. For the $\text{NaYF}_4@\text{SiO}_2$ NCs an additional peak for Si is observed and a strongly increased peak corresponding to O, as expected for silica overgrowth. The calculated stoichiometry for the NaYF_4 NCs, $\text{Na}_1\text{Y}_{0.6}\text{Er}_{0.02}\text{Yb}_{0.08}\text{F}_{3.2}$, is in fair agreement with the expected stoichiometry of $\text{Na}_1\text{Y}_{0.8}\text{Er}_{0.02}\text{Yb}_{0.18}\text{F}_4$.

Figures S1c and d (SI) show that the NCs have a hexagonal crystal structure (JCPDS No. 00-028-1192, cyan lines), as expected. Using the Scherrer equation,³⁷ a rough estimate of the core crystallite size can be calculated, which was *ca.* 18 nm and 17 nm for the NaYF_4 and $\text{NaYF}_4@\text{SiO}_2$ NCs

respectively. For the NaYF₄@SiO₂ NCs an extra broad band is present centered at a diffraction angle of *ca.* 28°, which is due to the amorphous silica.

By combining the data from TEM and EDX it was shown that NaYF₄ NCs doped with Er and Yb can be synthesized. The crystallite size from XRD and the NP size from TEM match very closely, indicating monocrystalline core NCs. Furthermore, silica overgrowth results in monodisperse NaYF₄@SiO₂ NCs in which the size, shape and crystal phase of the core NCs are unaltered.

Luminescence Properties. To investigate the luminescence properties of the NCs, room temperature emission and decay measurements were performed as shown in Figure 2. All measurements were performed using a continuous wave and a pulsed laser at 980 nm for the emission and decay measurements, respectively.

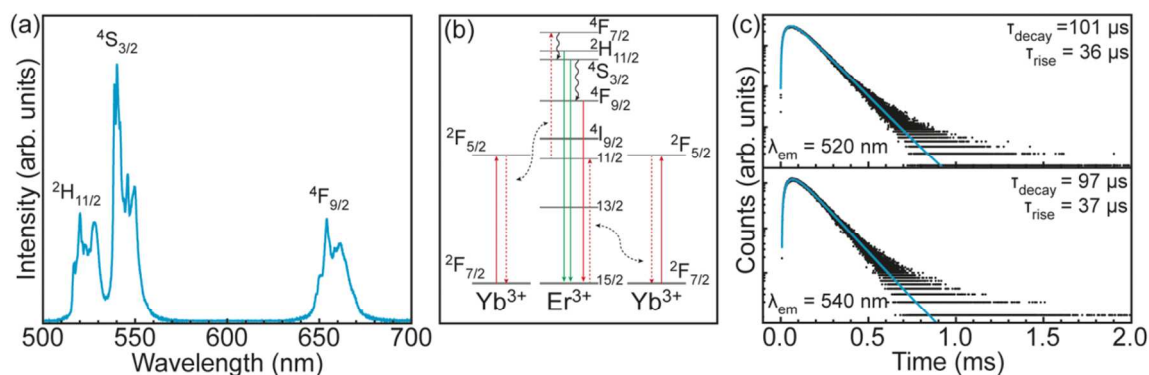


Figure 2. Emission spectrum of NaYF₄ NCs between 500 nm and 700 nm while exciting Yb³⁺ at 980 nm (a). The energy diagram of Er³⁺ and Yb³⁺, including the transitions involved in the upconversion process (b). The time-resolved luminescence measurements of emission from the ²H_{11/2} (520 nm) and ⁴S_{3/2} (545 nm) excited states upon excitation at 980 nm (c).

Figure 2a shows the upconversion (UC) emission spectrum upon excitation of Yb³⁺ at 980 nm. In the region between 500 nm and 700 nm three distinct peaks are observed, centered around 525 nm, 545 nm and 660 nm. These peaks correspond to the ²H_{11/2}-⁴I_{15/2}, ⁴S_{3/2}-⁴I_{15/2} and ⁴F_{9/2}-⁴I_{15/2} transitions of Er³⁺ respectively, as shown in Figure 2b.

Figure 2c depicts the luminescence lifetime measurements of the ²H_{11/2} (top) and ⁴S_{3/2} (bottom) excited state. The experimental data is shown in black, while a bi-exponential fit is shown in cyan. In

both cases, a rise and a subsequent decay in luminescence is observed. The rise times are *ca.* 35 μ s and can be explained by the upconversion energy transfer from Yb^{3+} to Er^{3+} . The decay time of *ca.* 100 μ s is determined by both radiative decay to all lower levels and non-radiative relaxation to the $^4\text{F}_{9/2}$ level of Er^{3+} , in agreement with earlier reports.³⁸ The luminescence decay curves of the $^2\text{H}_{11/2}$ ($\lambda_{\text{em}} = 520$ nm) and $^4\text{S}_{3/2}$ ($\lambda_{\text{em}} = 540$ nm) excited state are nearly identical which is expected for excited states that are thermally coupled.

Fluorescence Intensity Ratio Nanothermometry. FIR nanothermometry with luminescent nanoparticles relies on measuring the intensity ratio of emission lines from thermally coupled excited states. The population ratio of two excited states is determined by a Boltzmann distribution, which includes a temperature dependent factor, as shown in equation 1.

$$\frac{N_2}{N_1} = e^{-\frac{\Delta E}{kT}} \quad (1)$$

Here, ΔE is the energy difference between the two excited states, k is the Boltzmann constant and N_i is the population of state i . Emission intensities scale linearly with the population of an emitting state. This results in a temperature dependence of the luminescence intensity ratio as shown in equation 2.

$$\frac{I_2}{I_1} = \frac{g_2 A_2 h \nu_2 N_2}{g_1 A_1 h \nu_1 N_1} = C e^{-\frac{\Delta E}{kT}} \quad (2)$$

Here, I_i , g_i , A_i and ν_i are the (integrated) emission intensity, the degeneracy of the emitting level, the spontaneous emission rate and the frequency of the transition from state i to the ground state. For Er^{3+} in NaYF_4 NCs, the energy difference between the $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ excited states is *ca.* 711 cm^{-1} , as calculated from the spectrum in Figure 2a (using equation S1, SI).

To investigate the NCs as temperature probes, temperature dependent luminescence measurements were performed on powders of NCs at temperatures ranging from 300 - 600 K with intervals of 25 K. Each temperature interval consisted of 1 minute of heating, followed by a dwell time of at least 15 minutes before measuring spectra to ensure that the sample temperature was the same as that recorded by the thermocouple in the heating stage. Furthermore, the thermocouple allows for monitoring and compensation of heating effects due to the laser heating.

Figure 3a shows spectra, recorded from 300 K (black) up to 600 K (orange). The spectra are normalized to the maximum intensity of the $^4S_{3/2}$ - $^4I_{15/2}$ emission peak. Upon excitation at 980 nm, the characteristic luminescence peaks from the $^2H_{11/2}$ - $^4I_{15/2}$ and the $^4S_{3/2}$ - $^4I_{15/2}$ transitions can be observed at all temperatures. The energy difference between these peaks is calculated to be 710 cm^{-1} . The intensity ratio between these peaks changes with increasing temperature as expected for thermally coupled excited states.

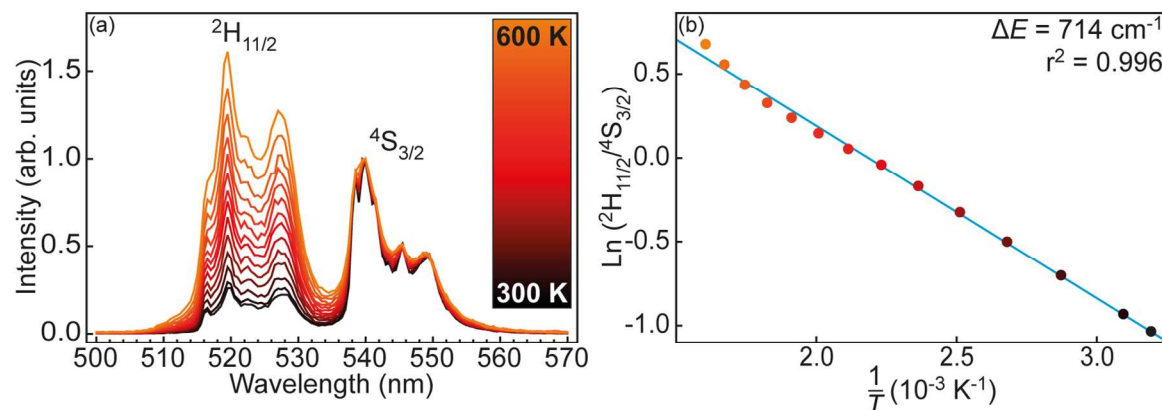


Figure 3. Emission spectra of NaYF₄ NCs upon excitation at 980 nm for temperatures ranging from 300 - 600 K recorded with steps of 25 K (a). The linear relation between the natural logarithm of the integrated intensity ratio of the $^2H_{11/2}$ and $^4S_{3/2}$ peak vs $1/T$ (b). The r-squared value for the fit, 0.996, is close to unity and the steepness of the fit corresponds to a ΔE of 714 cm^{-1} . Five spectra were measured at every temperature interval to estimate the error in the measurements; this error is within the dot size of the data points.

To evaluate the temperature-dependent luminescence further, the natural logarithm of the integrated intensity ratio is plotted versus $1/T$, as shown in Figure 3b. The data points can be fitted with a linear fit, as expected from equation 2. The quality of this fit is high, as shown by the r-squared value of 0.996. From the slope of the linear fit, the energy difference between the excited states, ΔE , is calculated to be 714 cm^{-1} in these NCs. This calculated ΔE closely matches the experimental values of ΔE of 711 cm^{-1} and 710 cm^{-1} determined from Figure 2a and 3a respectively.

Using the relation between temperature and luminescence intensity ratio of the $^4S_{3/2}$ and the $^2H_{11/2}$ peaks in Figure 3, it is evident that these NCs can be used as temperature probes from 300-600 K. At higher temperatures the overall luminescence intensity dropped and was too low to determine temperatures accurately. The intensity drop above 600 K is accompanied by a brown discoloration of the NC powders. The discoloration is most likely due to the oxidation of the organic ligands capping the NCs. Due to this decrease in luminescence intensity, the NCs cannot be used above 600 K for temperature sensing.

***In Situ* structural characterization.** *In Situ* TEM and XRD measurements were performed in order to investigate the origin of the sudden drop in luminescence intensity above 600 K. TEM micrographs were taken *in situ* while heating the NCs which were dropcasted on a special TEM heating chip with intervals of 50 K up to 875 K. XRD measurements, taking *ca.* 12 min for one XRD pattern, were performed continuously, while heating the sample to 825 K in 9 h and afterward cooling down in 45 min.

Figure 4 shows the resulting TEM images of core NCs at 475 K, 625 K and 775 K (a-c, respectively). Figure 4d shows three diffractograms taken before heating, during heating and after heating (top, middle and bottom, respectively). The FWHM and peak position of the (201) diffraction peak at 51° were extracted from all diffractograms and plotted versus temperature as shown in Figure 4e (top and bottom, respectively).

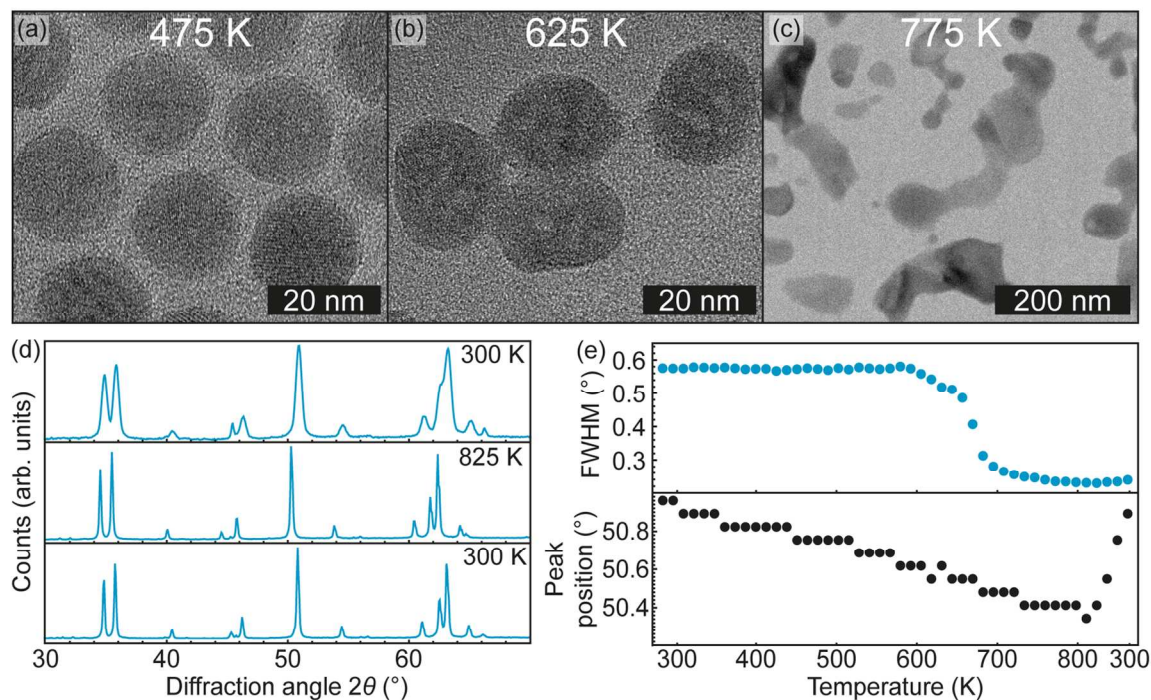


Figure 4. *In situ* TEM micrographs of NaYF₄ NCs at 475 K, 625 K and 775 K (a, b and c respectively). *In situ* XRD diffractograms (d) of NaYF₄ NCs before heating, during heating and after heating to 825 K (top, middle and bottom, respectively) and the corresponding FWHM and peak position of the (201) diffraction peak at 51° vs temperature (e).

Up to 575 K individual NCs can be observed in the TEM images. However, upon further heating, particles start to coalesce and subsequently melt together completely, forming large clusters of material. The temperature at which the coalescence starts, *i.e.* 625 K, matches the temperature at which the decrease in luminescence starts to occur. The XRD diffractograms show the characteristic peaks of hexagonal NaYF₄ shifting slightly upon heating due to thermal expansion of the crystal lattice. No phase transitions are observed, which is in accordance with investigations of Mathews and coworkers.³⁹ However, the diffraction peaks become irreversibly sharper around 625 K. The FWHM decreases from 0.6° to 0.2°, corresponding to an increase in crystallite size. To summarize, the results show that the NCs coalesce, starting at temperatures around 625 K.

The temperature at which the luminescence intensity decreases dramatically, corresponds to the temperature at which the NCs start to coalesce as confirmed with TEM and XRD. It is likely that the atomic alignment of the NCs during coalescence/sintering is imperfect, resulting in the introduction of

defects in the crystals. These defects can act as quenching sites for luminescence and therefore explain the sudden decrease in luminescence intensity in addition to the ligand oxidation discussed above.

The same *in situ* measurements were performed on the NaYF₄@SiO₂ NCs as shown in Figure 5. In contrast to the NaYF₄ NCs, the NaYF₄@SiO₂ NCs remain well separated on the TEM images for all temperatures showing that no interparticle alloying or intraparticle coalescence occurs. The FWHM of the (201) diffraction peak at 51° is constant during the entire thermal cycle, suggesting that the crystallite size remains unaltered. The overgrowth with silica, resulting in the core/shell architecture, ensures that the NCs are thermally stable up to 875 K and show no luminescence quenching up to at least 900 K.

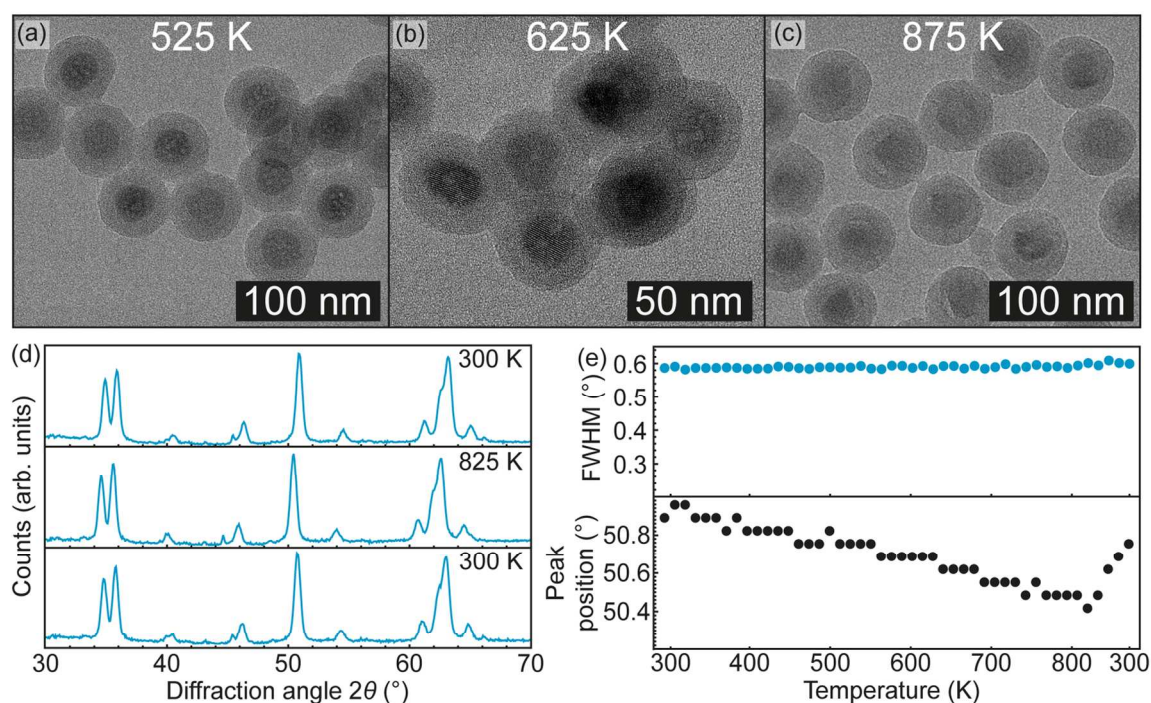


Figure 5. *In situ* TEM micrographs of NaYF₄@SiO₂ NCs at 525 K, 625 K and 875 K (a,b and c, respectively). *In situ* XRD diffractograms (d) of NaYF₄@SiO₂ NCs before heating, during heating and after heating to 825 K (top, middle and bottom, respectively) and the corresponding FWHM and peak position of the diffraction peak at 51° vs temperature (e).

(Temperature dependent) photoluminescence of NaYF₄@SiO₂ nanocrystals. To improve the temperature stability and thus extend the temperature range for nanothermometry, the NaYF₄ NCs were coated with a protective silica shell. The SiO₂ shell replaces the organic ligands and can prevent

luminescence quenching due to oxidation of ligands and NCs coalescence. To investigate if the silica overgrowth has an effect on the luminescence properties of the NC cores, (time-resolved) room temperature luminescence measurements were performed similar to those shown in Figure 2. The results, included in Figure S2, reveal that the emission spectra are nearly identical except for a small increase in the relative intensity of the $^4F_{9/2}$ - $^4I_{15/2}$ emission. Also the decay time of the $^4S_{3/2}$ and $^2H_{11/2}$ -excited states are slightly shorter (85 μ s after silica overgrowth *vs* 100 μ s before). The shortening of the lifetime suggests that non-radiative relaxation to lower lying $^4F_{9/2}$ states has increased, which is consistent with the observed increase in relative intensity of the $^4F_{9/2}$ - $^4I_{15/2}$ emission. The increase of non-radiative relaxation can be explained by the presence of hydroxyl groups in the silica shell.^{40,41} These hydroxyl groups are present because of incomplete condensation of neighboring silanol groups, Si-OH. These silanol groups condense to form typical Si-O-Si bonds in silica. The high energy O-H vibration (*ca.* 3500 cm^{-1}) gives rise to faster multiphonon relaxation.

Although the additional O-H vibrations gives rise to faster decay times as a result of multi-phonon relaxation, the energy levels of Er^{3+} remain unaltered due to the shielding of the f-orbitals. The unaltered energy levels dictate that the energy difference, ΔE , between the $^2H_{11/2}$ and $^4S_{3/2}$ state, and therefore the temperature dependent luminescence, is not expected to change which is in agreement with the experimental results.

In order to investigate the temperature-dependent luminescence properties of the silica coated NCs, emission spectra were measured at different temperatures under 980 nm excitation. Figure 6a shows the spectra taken from 300 K (black) up to 900 K (orange), normalized to the maximum intensity of the $^4S_{3/2}$ - $^4I_{15/2}$ emission peak. The characteristic luminescence peaks from the $^2H_{11/2}$ - $^4I_{15/2}$ and the $^4S_{3/2}$ - $^4I_{15/2}$ transitions can be observed up to 900 K!

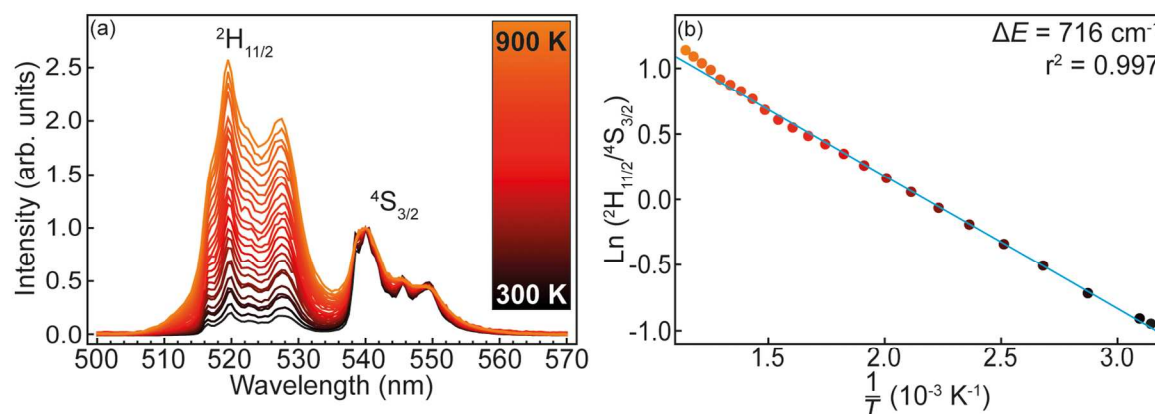


Figure 6. Emission spectra of NaYF₄@SiO₂ NCs upon excitation at 980 nm taken at temperatures ranging from 300 - 900 K with steps of 25 K (a). The linear relation between natural logarithm of the integrated intensity ratio of the ²H_{11/2} and ⁴S_{3/2} peak is plotted vs 1/T (b). The r-squared value for the fit, 0.997, is close to unity and the steepness of the fit corresponds to a ΔE of 716 cm⁻¹. Five spectra were measured at every temperature interval to estimate the error in the measurements; this error is within the dot size of the data points.

There is a small decrease in intensity but clearly the temperature stability of the emission is strongly improved in comparison to the NaYF₄ cores. At the temperature limit of our experimental set-up (900 K), the emission spectra are still well-resolved, suggesting that significantly higher temperatures can be probed. The emission intensity and signal to noise ratio hardly change between room temperature and 900 K. As can be observed in Figure 6a the signal-to-ratio remains high up to the highest temperatures. The temperature limit of 900 K is solely set by the limitations of our instrumentation.

To calibrate the silica coated NCs as a temperature probe, the natural logarithm of the integrated intensity ratio is plotted versus 1/T as shown in Figure 6b. The data points can be fitted with a linear fit, similar to Figure 3b.

The quality of the fit is high, as shown by the r-squared value of 0.997. From the slope of the linear fit, the energy difference between the excited states, ΔE , can be calculated to be 716 cm⁻¹ and closely matches the experimental value of ΔE determined from the spectra and from the temperature dependent emission spectra of the NaYF₄ core particles. The results show that the maximum temperature range of

the core particles can be expanded up to at least 900 K by applying a protective silica shell. The linear fit shows good agreement with the data points over the full temperature range.

Thermal cycling experiment. In addition to having a high thermal stability and temperature accuracy, it is important that a temperature sensor can survive multiple thermal cycles without change in properties. To investigate the durability of the $\text{NaYF}_4@\text{SiO}_2$ NCs, the temperature probes were cycled between 300 K and 900 K five times. During the heating up and cooling down in each cycle luminescence measurements were performed at 300 K, 450 K, 600 K, 750 K and 900 K.

Figure 7a shows the spectra taken at five different temperatures during the cycle experiment, normalized to the $^4\text{S}_{3/2}$ - $^4\text{I}_{15/2}$ emission peak. The spectra at a specific temperature are very similar. At each temperature the intensity ratio of the two emission peaks was determined and plotted in Figure 7b. The data points (black) follow identical trends during the five cycles (cyan line). The average fluorescence intensity ratios for each temperature were calculated from the data (dashed lines). The data points only have small deviations from the average. The thermal cycling experiment shows that the temperature probes are stable and durable up to 900 K.

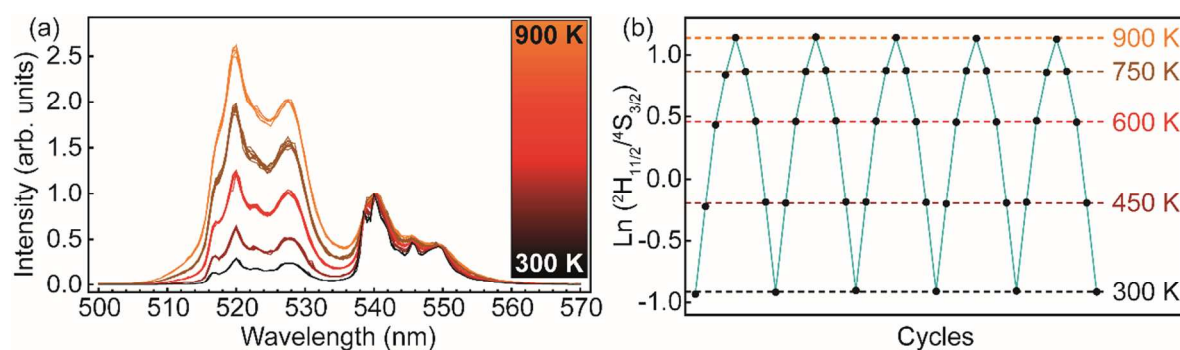


Figure 7. Emission spectra of $\text{NaYF}_4@\text{SiO}_2$ upon excitation at 980 nm while cycling the temperature between 300 K, 450 K, 600 K, 750 K and 900 K for 5 times (a). The natural logarithm of the fluorescence intensity ratio plotted per cycle step (black dots). The cyan line is for directing the eyes and the dashed lines are the means of the data points per temperature (b).

The standard deviations in temperature are calculated to be 1.1 K, 1.8 K, 2.9 K, 5.2 K and 5.3 K for 300 K, 450 K, 600 K, 750 K and 900 K, respectively. These values are high compared to the very

accurate temperature probes used for bio-imaging ($< 0.5 \text{ K}^{15}$). Temperature accuracies down to 0.2 or 0.3 K have been reported for the Yb-Er and Yb-Tm couple in the biologically interesting temperature window.⁴⁶⁻⁴⁹ The smaller standard deviations for temperature measurements in bio-imaging are expected based on the smaller temperature range. The probes used are optimized for this temperature range such that the change in intensity ratio with temperature is large.

In the case of FIRT, also the sensitivity S and relative sensitivity of a temperature probe are often used as a quality measure of the performance as a temperature sensor. S is defined as the change of the intensity ratio R with temperature, as shown in equation 3.

$$S = \frac{dR}{dT} \quad (3)$$

The relative sensitivity is defined as S/R (in $\% \text{ K}^{-1}$). For a large temperature range, the change in the intensity ratio with temperature varies with temperature and thus the relative sensitivity will be different in different temperature regimes. For the $\text{NaYF}_4\text{:Er,Yb}$ nanoparticles in the 300 to 900 K temperature range the sensitivity, calculated using equation 2 and 3 (with $C=9.4$), shows that the relative sensitivity decreases from $1.02\% \text{ K}^{-1}$ at 300 K to $0.13\% \text{ K}^{-1}$ at 900 K. The $1.02\% \text{ K}^{-1}$ sensitivity is consistent with values typically found around 300 K for Er^{3+} (between 0.5 and $1.5\% \text{ K}^{-1}$).⁵⁰ The variation in relative sensitivity with temperature is inherent to using temperature probes in a wide range. The sensitivity of a temperature probe cannot be optimized for the full temperature window. Most applications which require a broader temperature sensing range do not require a sensitivity of 0.5 K. However, care should be taken that the error in the temperature measurement does not become too large. In our case a measurement error of *ca.* 5 K at 900 K is sufficient for most applications.

The present results demonstrate for the first time the use of *ca.* 50 nm NCs for high temperature nanothermometry using a silica coating to stabilize the NCs. Silica-stabilized NCs of other host materials (*e.g.* oxides) and also other Ln^{3+} dopants with a different ΔE between levels can be developed

to realize other high temperature nanoprobe with potentially higher sensitivity in the high temperature range.⁵¹

In summary, we have reported NaYF₄:Er³⁺,Yb³⁺ nanocrystals coated with a SiO₂ shell as stable nanoprobe for high temperature nanothermometry, up to 900 K. The temperature-dependent intensity ratio of emission from the ²H_{11/2} level (520 nm) and the ⁴S_{3/2} level (545 nm) follows a Boltzmann distribution. From the intensity ratio temperatures can be measured accurately (1 to 5 K accuracy) using remote sensing techniques. The temperature stability of the probe is confirmed by *in situ* TEM and XRD. The temperature-dependent luminescence remains unaltered in multiple cycling experiments between 300 K and 900 K. Based on these observations, the NaYF₄:Er³⁺,Yb³⁺ nanoprobe provide a non-invasive temperature probe that is dispersible in liquids and has a broad range for temperature sensing. It can therefore be used for high resolution temperature sensing in a wide range of applications, including chemical reactors and nano-electronics. The reported method of strongly improving the thermal stability of nanoprobe by silica coating can be generally applied to enhance the performance of other nanoprobe.

Supporting Information. The experimental methods are available as supporting information. EDX and *ex situ* XRD data for both NaYF₄ and NaYF₄@SiO₂ is shown in Figure S1. (Time resolved) luminescence measurements of the NaYF₄@SiO₂ NCs is shown in Figure S2. This material is available free of charge via the Internet on the ACS publication website.

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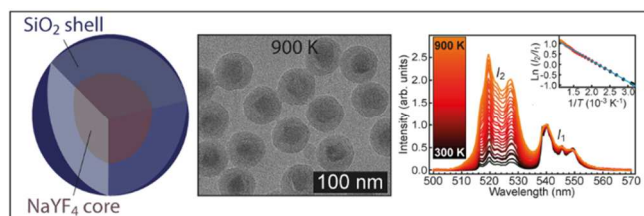
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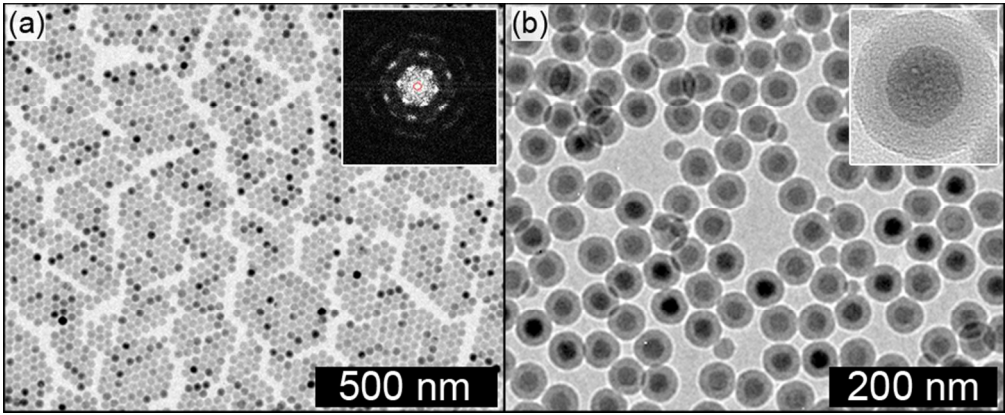


Figure 1. TEM micrographs of the prepared NaYF₄ NCs (a, including inset with a Fourier Transform) and NaYF₄@SiO₂ NCs (b). The average diameter of the core NCs is 21.7 ± 1.1 nm and the silica shell is 10.0 ± 0.7 nm.
Figure 1
85x34mm (300 x 300 DPI)

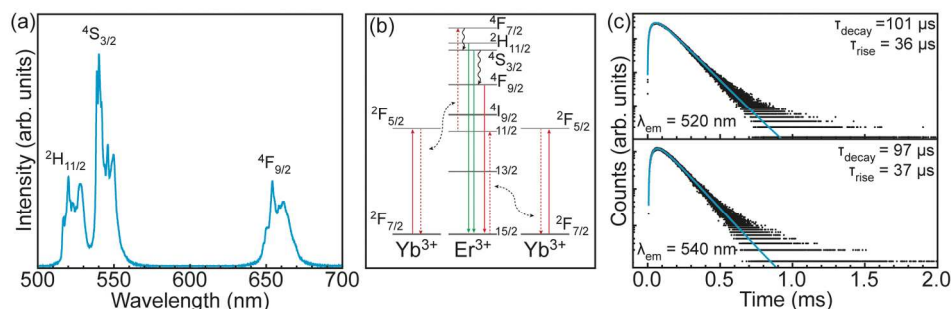


Figure 2. Emission spectrum of NaYF₄ NCs between 500 nm and 700 nm while exciting Yb³⁺ at 980 nm (a). The energy diagram of Er³⁺ and Yb³⁺, including the transitions involved in the upconversion process (b). The time-resolved luminescence measurements of emission from the ²H_{11/2} (520 nm) and ⁴S_{3/2} (545 nm) excited states upon excitation at 980 nm (c).

Figure 2

160x50mm (300 x 300 DPI)

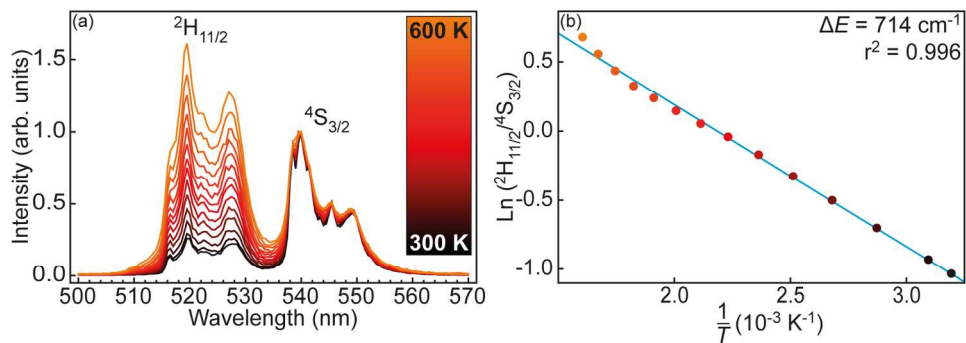


Figure 3. Emission spectra of NaYF₄ NCs upon excitation at 980 nm for temperatures ranging from 300 - 600 K recorded with steps of 25 K (a). The linear relation between the natural logarithm of the integrated intensity ratio of the ²H_{11/2} and ⁴S_{3/2} peak vs 1/T (b). The r-squared value for the fit, 0.996, is close to unity and the steepness of the fit corresponds to a ΔE of 714 cm⁻¹. Five spectra were measured at every temperature interval to estimate the error in the measurements; this error is within the dot size of the data points.

Figure 3
159x56mm (300 x 300 DPI)

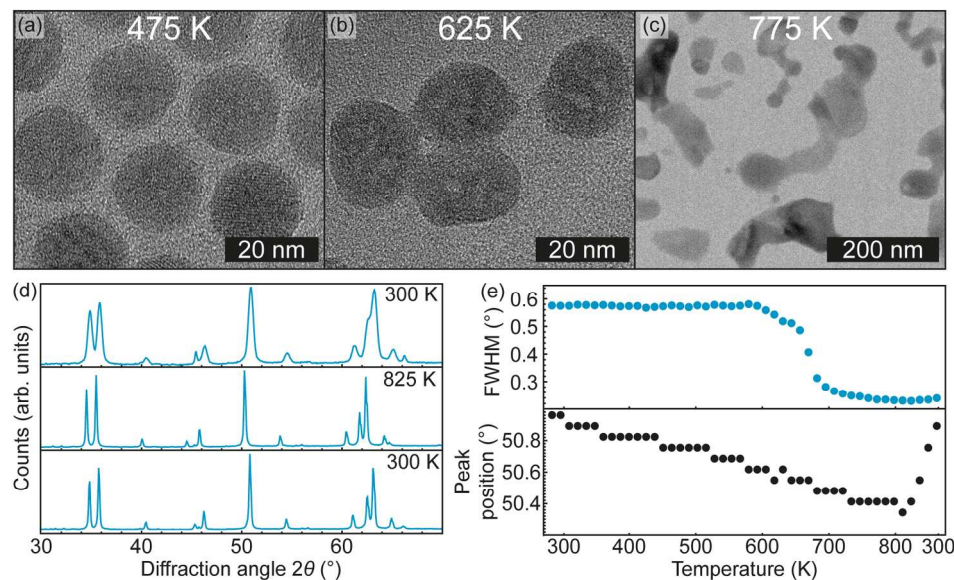


Figure 4. *In situ* TEM micrographs of NaYF₄ NCs at 475 K, 625 K and 775 K (a, b and c respectively). *In situ* XRD diffractograms (d) of NaYF₄ NCs before heating, during heating and after heating to 825 K (top, middle and bottom, respectively) and the corresponding FWHM and peak position of the (201) diffraction peak at 51° vs temperature (e).

Figure 4

160x93mm (300 x 300 DPI)

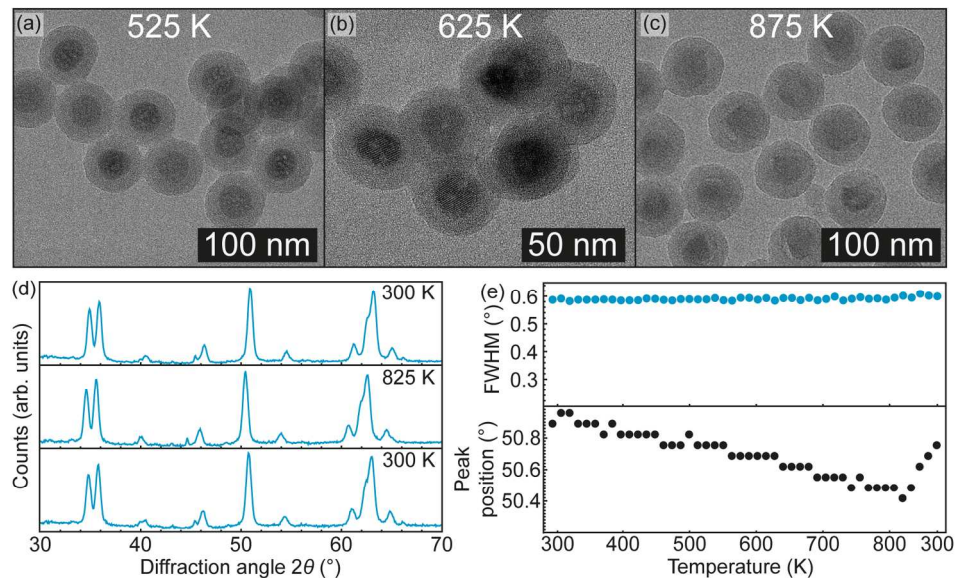


Figure 5. *In situ* TEM micrographs of NaYF₄@SiO₂ NCs at 525 K, 625 K and 875 K (a,b and c, respectively). *In situ* XRD diffractograms (d) of NaYF₄@SiO₂ NCs before heating, during heating and after heating to 825 K (top, middle and bottom, respectively) and the corresponding FWHM and peak position of the diffraction peak at 51° vs temperature (e).

Figure 5
159x93mm (300 x 300 DPI)

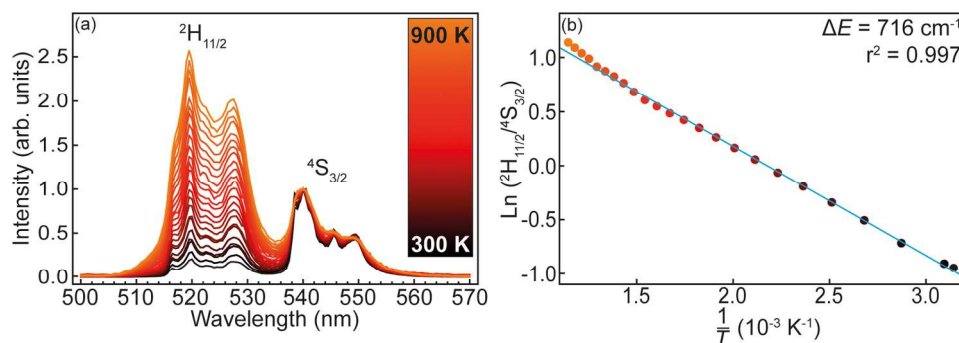


Figure 6. Emission spectra of NaYF₄@SiO₂ NCs upon excitation at 980 nm taken at temperatures ranging from 300 - 900 K with steps of 25 K (a). The linear relation between natural logarithm of the integrated intensity ratio of the ²H_{11/2} and ⁴S_{3/2} peak is plotted vs 1/T (b). The r-squared value for the fit, 0.997, is close to unity and the steepness of the fit corresponds to a ΔE of 716 cm⁻¹. Five spectra were measured at every temperature interval to estimate the error in the measurements; this error is within the dot size of the data points.

Figure 6
159x56mm (300 x 300 DPI)

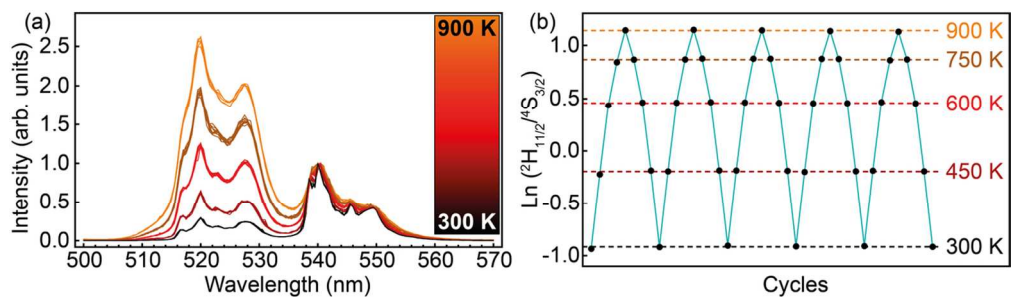


Figure 7. Emission spectra of NaYF4@SiO2 upon excitation at 980 nm while cycling the temperature between 300 K, 450 K, 600 K, 750 K and 900 K for 5 times (a). The natural logarithm of the fluorescence intensity ratio plotted per cycle step (black dots). The cyan line is for directing the eyes and the dashed lines are the means of the data points per temperature (b).

Figure 7
116x33mm (300 x 300 DPI)